

OLEDs for Automotive Lighting

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Abstract

In this work we describe automotive OLED lighting panel development for various functions. New red-emitting panels with high brightness and long operational stability for turn, stop and V2X communication are described. High brightness, high efficiency cyan-emitting panels have been demonstrated for self-driving car modes. Highly segmented bendable panels for curved surfaces are presented.

Author Keywords

OLED; automotive lighting; stacked OLED; efficacy; thermal stability; V2X; cyan; bendable.

1. Objective and Background

Automotive lighting technology has been evolving from simple road illumination for driving safety into functional, specialized, and customizable lighting solutions. Automotive manufacturers are increasingly integrating OLED panels into their vehicles. Currently, red panels are used in taillight functions with plans for higher brightness panels to be incorporated for vehicle-to-everything (V2X) communication, turn and stop signals. High brightness cyan, amber and white OLED panels are also being developed for self-driving modes, turn indicators and lit logos. The thinness of OLEDs facilitates their seamless integration into compact spaces and, with bendable panels, enables lighting of curved surfaces. The intrinsic capability to combine segmentation and uniform lit surfaces creates opportunities for unique branding and personalization. OLED automotive lighting technology paves the way for new form factors, new functional uses and superior safety systems. [1]



Figure 1. Atala Wave segmented automotive OLED panels shown at CES2025.

2. Results

OLEDWorks' Atala automotive OLED technology includes many attributes, such as stacked device architecture with charge generation layers for high luminance and long lifetime, proprietary thin film encapsulation, metallization layer on top of the ITO for current distribution and improved uniformity and rigid or flexible substrates. For red OLED lighting panels there are strict color requirements. The OLED panels must have CIE_{x,y} coordinates of (0.7, 0.3) or deeper red, dominant wavelength >627 nm and minimal color change with viewing angle. The panels also must meet automotive reliability standards. Environmental testing includes high temperature (85°C and 105°C), high humidity (85°C/85%RH), sun light exposure and other tests. Typical temperature conditions for automotive exterior applications are between -40°C to 105°C. [2] Performance of commercial red panels is shown in Table 1.

Table 1: Performance of commercial red taillight OLEDs

Brightness @ 25°C	2,000 cd/ m2
Peak wavelength	643 nm
Dominant wavelength	627 nm
CIE _{x,y}	0.703, 0.297
Operating Lifetime at 60°C@ 2000 cd/m ²	20,000 hours
Storage lifetime	15 years
OLED Stacks	2
Panel thickness	0.9 mm

Table 2: Performance of next generation of high brightness red OLEDs. Data at 13 mA/cm².

Stacks	V	cd/m2	cd/A	CIE _x	CIE _y	EQE, %	Dom Wavelength, nm
2	6.4	3175	24.4	0.707	0.293	45.1	629
4	12.3	6715	51.7	0.706	0.293	86.4	629
6	18.0	10670	82.1	0.704	0.295	118	628

Brightness of the commercial panels is 2,000 cd/m² (Table 1) but the next generation of red OLEDs must produce luminance in the range of 5,000 - 15,000 cd/m². To achieve the required high luminance from a panel, we have developed an improved OLED formulation and, for some applications, increased the number of stacks. However, multi-stack OLEDs exhibit significant self-heating that

might accelerate degradation processes. To minimize such effects each stack is optimized on thermal stability.

To better understand the mechanisms of OLED degradation when the device is operated at elevated temperatures, we developed a fast stability testing procedure. In this test, the OLED panel is fixed on a hotplate (T_{hotplate} can be varied from 65°C to 105 °C) via thermally conductive adhesive tape. OLED emission intensity is monitored as a function of time with the photodetector situated about 2 cm above the panel to minimize heat transfer.

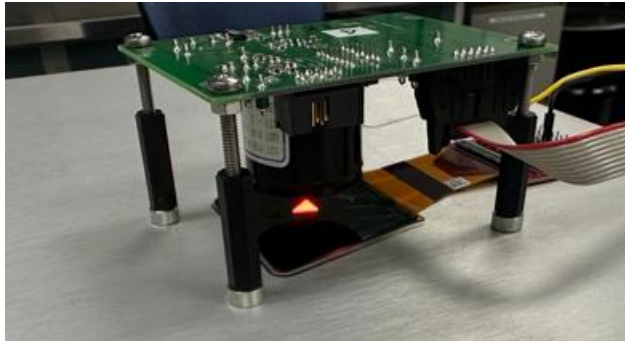


Figure 2. Thermal stability test with single segment on. The segment is driven at 26 mA/cm².

The results of the stability tests at 105°C and at 26 mA/cm² are shown in Figure 3. The 2 stack and 4 stack red OLEDs age significantly differently under these conditions. These data show that luminance decays more slowly in 4 stack OLEDs compared to 2 stack devices despite higher electrical power during operation. Because both types of devices age at the same constant current, we can conclude that 4 stack devices exhibit higher thermal stability compared to the commercial 2 stack panels. Thus, the 4 stack devices can be used in applications at higher brightness while still providing the required long lifetime.

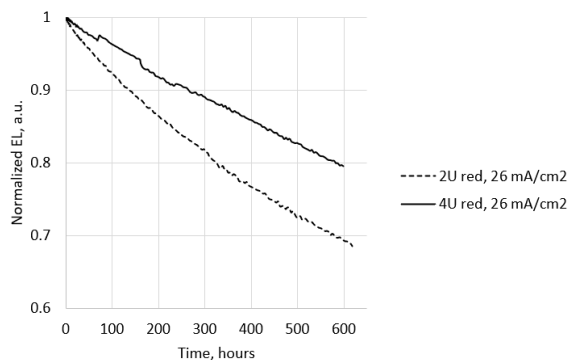


Figure 3. Normalized light output as a function of aging time for 2 stack (2U) and 4 stack (4U) red OLEDs at 105°C and 26 mA/cm².

Cyan illumination is an emerging technology in automotive lighting for certain driver assistance modes and for fully autonomous cars. This technology allows for seamless communication between the vehicle and its surroundings and is employed to communicate a switch from human to system control. With the number of vehicles with advanced driver assistance modes and with the number of autonomous vehicles on the rise, it is crucial to develop a reliable cyan lighting technology that complies with the brightness, efficiency, color, and stability specifications defined by automotive standards and car manufacturers. OLEDWorks has utilized its multi-stack OLED technology to develop high-brightness (>10,000 cd/m² at 10 mA/cm²), high-efficiency (>50% EQE at 10,000 cd/m²) cyan OLED panels.

To predict accurately performance of a multi-stack cyan OLED, it is important to understand the recombination zone (RZ) position within the stack. This problem is commonly solved by introducing a low-energy probe/sensitizer in different locations throughout the emissive layer (EML) and tracking the sensitizer emission strength as a function of its location. The spatial resolution of such method is dictated by Förster energy transfer radius between the emitter and sensitizer, which can be on the order of 1-5 nm, depending on the exact donor-acceptor pairing [3]. However, the shortcoming is the assumption that the sensitizer itself has no impact on charge transport and exciton density distribution. Herein, a sensitizer-free method with lower spatial accuracy (5-10 nm) is employed to get an understanding of the RZ position in cyan OLEDs. The advantage of such method is simplicity in data interpretation, as only 5 different devices with unaltered emissive layer concentrations are required to draw conclusions about the RZ position.

These single-stack devices are shown in **Error! Reference source not found.** 4 (a), where the device of interest with EML thickness of 40 nm is compared to four control devices 1-4 with 10 nm EML each placed at certain regions within the EML. Additionally, emitter location within the cavity in control devices is chosen to be close to the optical node. Contrary to the common practice of placing an emitter in the optical anti-node to maximize emission intensity, such choice is motivated by obtaining control device spectra with significant variations in both shape (**Error! Reference source not found.**(b)) and intensity (**Error! Reference source not found.**(c)). Device of interest spectrum, denoted by black line in **Error! Reference source not found.**(c), is subsequently compared to the control devices. Its shape and intensity closely match the “Device 2” control, suggesting that exciton recombination takes place not far (within 10 nm) from the EML center, but on the side closer to the EML/HBL interface. The spatial accuracy of this method can be further improved by having the control device EML thickness thinner than 10 nm. [4]

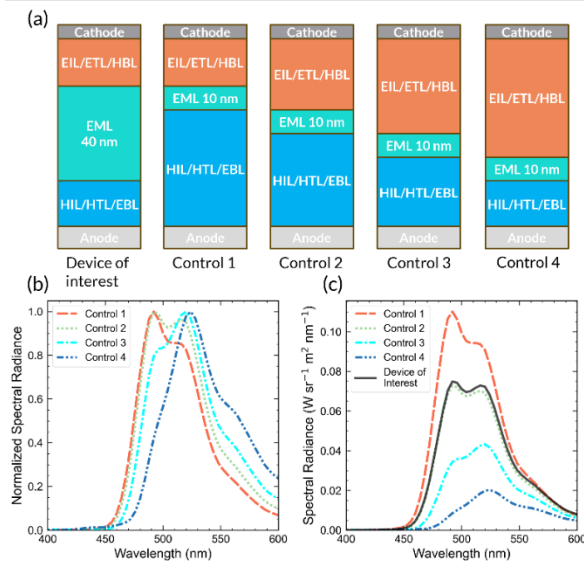


Figure 4. (a) Schematic illustration of the device of interest and 4 controls. All layers, except for the EML, are not drawn to scale. (b) Normalized spectral radiance of the 4 control devices. (c) Spectral radiance of the device of interest (solid black line) compared against the spectra of control devices.

Using the understanding of RZ position in the single-stack cyan OLED, a series of multi-stack cyan OLED panels were fabricated. The brightnesses of $>10,000\ cd/m^2$ and $>20,000\ cd/m^2$ are reached at current densities of $10\ mA/cm^2$ and $30\ mA/cm^2$, respectively, thus satisfying automotive requirements for both night and day-time illumination.

Beyond the performance of the OLED formulations in automotive lighting applications, the types and design of the substrates used to create our Atala automotive lighting panels are major factors in the benefits of OLED technology for automotive lighting. Initial applications used only rigid glass substrates (0.7 mm glass) with less than 10 separately addressable segments per panel. To expand the application space, thin bendable glass substrates and higher levels of segmentation (60-500 segments per panel) have been developed.

Use of flexible substrates for Atala Wave automotive panels allows light to follow vehicle curves for greater visibility and design freedom. Thin (0.1 mm) flexible Atala Wave OLED panels have been commercialized recently. These commercial bendable OLED lighting panels have been tested and validated at a minimum bending radius of 200 mm (see Figure 5). Panel performance is the same as that of our Atala Digital rigid glass OLEDs; the flexible panels passed all strict automotive tests and are designed to last 15+ years in exterior automotive use.

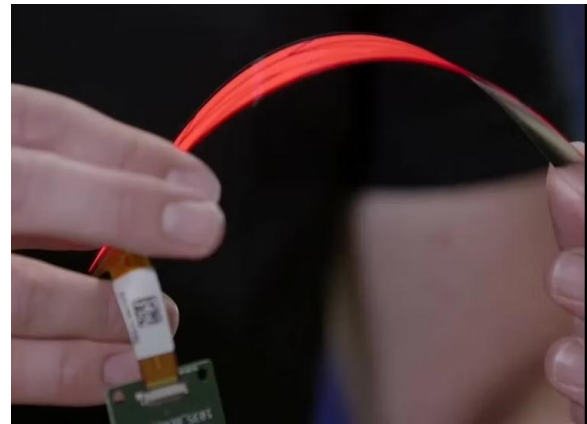


Figure 5. Atala Wave bendable panel with a 200 mm minimum bend radius.

Because of uniform emission profile, Atala OLED panels are well suited for enhanced communication through rendering of sharp icons and other symbols as well as dynamic lighting animations. Atala OLED panels can be divided into segments/pixels with sharp contrast edges and narrow/invisible gaps. Atala Digital OLED panels with up to 60 segments (see Figure 6) have been commercialized and panels with 500 and more segments have been demonstrated. OLED segment sizes as small as $3\ mm^2$ and unlit gaps as small as $40\ \mu m$ have also been demonstrated. Each segment is individually controlled, enabling each panel to display a variety of symbols. With these features

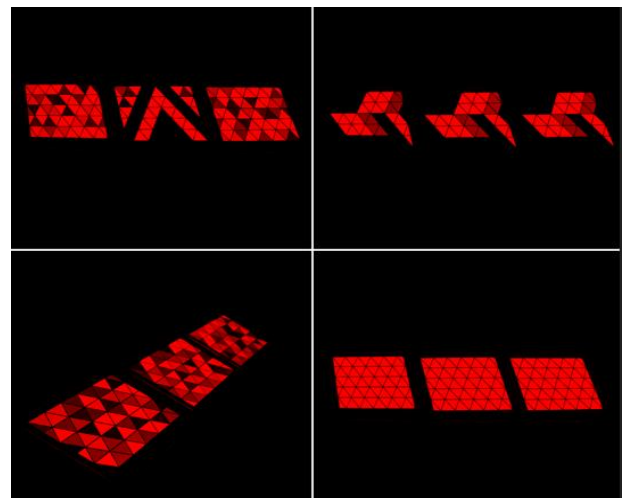


Figure 6. Atala Digital panels with 60 individually controlled segments

the Atala OLED panel becomes a highly pixelated automotive exterior light source/display. Dimming effects in combination with pixelation enables 3-D visual effects on a thin 2-D surface.

3. Impact

OLEDWorks is developing high brightness multi-stack Atala OLED panels qualified for automotive applications. The lighting technology enables designers to customize the size and shape of rigid and bendable automotive panels with custom designed segmentation (size, shape and count) and different luminance requirements for each location and function on the vehicle. Red-emitting devices with up to 15,000 cd/m² luminance for stop, turn, and daylight V2X communication have been demonstrated. Efficient high brightness cyan-emitting panels with >10,000 cd/m² have been developed for self-driving modes and autonomous car applications. High brightness amber- and white-emitting OLEDs are also under development. All panels undergo extensive automotive testing and quality control to meet automotive regulations and performance requirements. Product reliability is supported by an in-house environmental test laboratory and equipment. Both OLEDWorks sites -Aachen manufacturing and Rochester R&D sites are IATF16949 and ISO-9001 certified.

4. Acknowledgments

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5. References

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